

Ideas (in order of preference)

Idea 1: IoT Environmental Monitoring System

- **System Architecture:** The MSP430 microcontroller continuously samples ambient temperature and humidity metrics within a high-density server rack enclosure.
- **Cloud Telemetry & Processing:** Data is transmitted via MQTT to a remote cloud broker. The cloud infrastructure evaluates the environmental state against predefined safe operating thresholds to detect thermal anomalies or moisture risks (such as electrostatic discharge or condensation).
- **Automated Actuation:** Upon detecting critical environmental levels, the cloud issues a control payload back to the edge device. The MSP430 receives this command and dynamically adjusts the cooling fan speeds using Pulse Width Modulation (PWM).

Idea 2: Cloud-Integrated Liquid Leak Detector

- **System Architecture:** Designed for modern liquid-cooled data centres, the MSP430 interfaces with moisture sensors strategically placed near critical coolant lines or containment trays.
- **Cloud Telemetry & Processing:** Upon detecting anomalous moisture, the edge system transmits a high-priority alert to the cloud via MQTT. The cloud logs the fault for maintenance tracking and immediately dispatches an emergency override command.
- **Automated Actuation:** The MSP430 processes the incoming command and triggers an isolation relay, simulating the mechanical closure of a localized emergency water valve to halt fluid flow.

Idea 3: Smart Power Load Shedding System

- **System Architecture:** The MSP430 utilizes current sensors to continuously monitor the electrical power consumption of a simulated server rack infrastructure.
- **Cloud Telemetry & Processing:** Energy consumption metrics are forwarded to the cloud via MQTT. If the system detects a sustained power anomaly or a dangerous overload condition, the cloud processing unit calculates the required load reduction.
- **Automated Actuation:** A mitigation command is sent back to the MSP430, which triggers secondary relays to safely disconnect non-critical server loads, preserving power for priority systems.